

MATHEMATICS – QUESTION PAPER

Time : 1½ Hours

Total Marks : 25

Max. Marks : 25

1. Answer the following short questions. [5 × 5 = 25]

- (i) Using the forward difference operator, show that $\Delta^3 e^x = (e^h - 1)^3 e^x$. Hence find its value when $h = 1$.
- (ii) What is the degree of precision of the Trapezoidal Rule?
- (iii) Define extrapolation.
- (iv) Give an example of truncation error, with reference to the Maclaurin series expansion of e^x .
- (v) When is Newton's Forward Interpolation Formula used? (i.e., under what conditions on spacing and location of the required point)

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2. (i) Show that $\Delta \log f(x) = \log \left[1 + \frac{\Delta f(x)}{f(x)} \right]$. [5]

(ii) Show that $E^{\frac{1}{2}} = \mu + \frac{1}{2} \delta$, where μ is the mean operator and δ is the central difference operator.

3. Given the following table, two values of $f(x)$ are missing. [5]
Assuming $f(x)$ is a polynomial of degree 2, find the missing values :

x	0	1	2	3	4	5
$f(x)$	0	?	8	15	?	35

4. Using Lagrange's Interpolation Formula, find $f(5)$ from the following data : [5]

x	3	7	9	12
y	11	17	24	30

5. Evaluate the following integral using the Trapezoidal Rule [5]
(with the given number of subintervals) :

$\int_1 x e^x dx$ over the given limits (the sheet works through limits from -1 to 0, and again from 1 to 2, with $h = \frac{1}{6}$).